

# Hong Yuan Cao

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## EDUCATION

### Boston University

Boston, MA

Double Major: Bachelor of Arts in Computer Science and Economics, GPA: 3.74/4.0   Expected Graduation: May 2026

Relevant Courses: Distributed Systems, Databases, Software Engineering, Analysis of Algorithms, Computer Systems, Probability in Computing, Combinatoric Structures, Geometric Algorithms, Discrete Mathematics, Data Structures

## TECHNICAL SKILLS

**Programming Languages:** Python, JavaScript, TypeScript, Java, C#, Go, SQL, HTML/CSS, LaTeX, Assembly

**Databases & Cloud:** MySQL, PostgreSQL, Redis, AWS, Supabase, Alibaba Cloud

**Frameworks & Libraries:** Django, React.js, Vue.js, React Native, FastAPI, Electron, Flask, Node.js, Spring Boot, JUnit, pytest, PyTorch, NumPy, Pandas, OpenCV, Scikit-learn, NLTK, Matplotlib, BeautifulSoup, Axios, Terraform, Docker, Git

## EXPERIENCES

### Incoming Software Engineering Intern

Shanghai, China

KeelinMicro

Summer 2025

- Will be working on drivers for automotive vehicles with multi-sensor synchronization and precision timing protocols

### Software Engineering Intern

Shanghai, China

Tospur Real Estate Consulting Co., Ltd

June 2024 – August 2024

- Built **Django** ETL pipeline on **Alibaba Cloud** consolidating 4 **MySQL** databases, reducing aggregation by **45%**
- Implemented APIs with credential migration, successfully transitioning **200+** consultants from 4 separate systems
- Optimized database performance via **B-tree indexing** and **connection pooling**, reducing query latency by **23%**

### Machine Learning Intern

Shanghai, China

Anban Tech

May 2024 - June 2024

- Built preprocessing pipeline to clean code-test mappings for fuzzing test generation LLM using **Pandas** and **regex**
- Curated **2.2GB JUnit 5** dataset from MIT/Apache repositories using **Git API** for fuzzing LLM training
- Developed **Python** validation scripts achieving **95% standardization** of dataset formatting for training pipeline

### Software Engineering Intern

San Francisco, CA

Tonic – Sprout Technologies, Inc.

June 2023 – October 2023

- Implemented responsive UI components and animations using **React.js**, increasing landing page engagement by **24%**
- Built a social media crawler using **Python (BeautifulSoup4)** that identified **200+** qualified influencers to target
- Filtered **3,000+** media contacts using **Python (Pandas)** and Excel, improving outreach response rates by **25%**

## PROJECTS

### Athens - AI Learning Platform | FastAPI, React, TypeScript, OpenAI, Supabase, ChromaDB

- Built full-stack architecture with **FastAPI**, **React TypeScript**, and hybrid **ChromaDB/Supabase** storage
- Developed end-to-end **RAG** system with document ingestion, vector embeddings, and semantic retrieval for Q&A
- Implemented **prompt engineering** with context injection and memory, improving response quality by **40%**

### Serverless Image Processing Pipeline | AWS S3, Lambda, CloudFront, DynamoDB, Terraform, React

- Deployed serverless infrastructure using **Terraform**, **S3**, **Lambda**, **CloudFront**, reducing delivery latency by **38%**
- Implemented **Python**-based **Lambda** functions triggered by **S3** events to automatically resize uploaded images
- Built **React.js** frontend connecting to AWS services through **API Gateway**, storing metadata in **DynamoDB**

### Multi-Currency Budget Tracking Mobile App | Django, React Native, PostgreSQL, Android Studio

- Developed a expense tracking app with **React Native**, featuring currency conversion via third-party API integration
- Built **Django REST** APIs using **composite indexing** and **ORM** optimization, reducing retrieval by **65%**
- Implemented offline caching with **AsyncStorage**, improving response time from **1.2s to 400ms**

### Movie Recommendation Platform | Flask, Vue.js, Python, Javascript, PostgreSQL

- Integrated movie database API with **Flask** to create an interactive dashboard using **Vue.js** components
- Built content-based recommendation algorithm using **CountVectorizer** and **cosine similarity** to analyze metadata
- Created secure user authentication with **Flask-Login** and **bcrypt**, storing user data in **PostgreSQL**

### Neural Language Model Implementation | Python, PyTorch

- Implemented transformer architecture in **PyTorch** with token and positional embeddings for sequential prediction
- Designed custom attention mechanisms for next-token prediction, enabling model to generate coherent text
- Implemented training pipeline with **AdamW** optimizer and batch processing, reducing model perplexity by 30%